

# Phase 4 — Sync Expansion Implementation Plan

| **For agentic workers:** Use subagent-driven-development.

**Goal:** Add Sync Now buttons, device identity, history + credentials sync categories, expand Menu sync settings.

**Architecture:** Extend existing Settings, SyncPeriod, BackgroundService, WorkManager workers. Add generated UUID for device identity. No cloud backend needed.

**Tech Stack:** WorkManager, Room, Orbit MVI, SharedPreferences

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## Task 1: Add device identity to Settings

### Files:

- Modify: core/models/src/main/kotlin/lava/models/settings/Settings.kt
- Modify: core/preferences/src/main/kotlin/lava/securestorage/PreferencesStorageImpl.kt
- Modify: core/preferences/src/main/kotlin/lava/securestorage/PreferencesStorage.kt
- Modify: core/data/src/main/kotlin/lava/data/impl/repository/SettingsRepositoryImpl.kt

### Steps:

1. Add val deviceId: String = "" to Settings data class
  2. In PreferencesStorageImpl, add key "device\_id" — on first read, if empty/null, generate and persist UUID.randomUUID().toString()
  3. Add getDeviceId(): String to PreferencesStorage interface
  4. In SettingsRepositoryImpl, include deviceId in the settings flow
  5. Compile, spotless, commit
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## Task 2: Add history + credentials sync periods to Settings

### Files:

- Modify: core/models/src/main/kotlin/lava/models/settings/Settings.kt
- Modify: core/models/src/main/kotlin/lava/models/settings/SyncPeriod.kt (add THIRTY\_MINUTES)
- Modify: core/preferences/.../PreferencesStorageImpl.kt
- Modify: core/preferences/.../PreferencesStorage.kt
- Modify: core/data/.../SettingsRepositoryImpl.kt

### Steps:

1. Add `val historySyncPeriod: SyncPeriod = SyncPeriod.OFF` and `val credentialsSyncPeriod: SyncPeriod = SyncPeriod.OFF` to Settings
  2. Add THIRTY\_MINUTES to SyncPeriod enum
  3. Persist via SharedPreferences keys "history\_sync\_period" and "credentials\_sync\_period"
  4. Add repository methods `setHistorySyncPeriod()` and `setCredentialsSyncPeriod()`
  5. Compile, spotless, commit
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## Task 3: Add Sync Now button to Favorites + Bookmarks

### Files:

- Modify: feature/favorites/src/main/kotlin/lava/favorites/FavoritesScreen.kt
- Modify: feature/favorites/src/main/kotlin/lava/favorites/FavoritesViewModel.kt
- Modify: feature/favorites/src/main/kotlin/lava/favorites/FavoritesAction.kt
- Modify: feature/bookmarks/src/main/kotlin/lava/forum/bookmarks/BookmarksScreen.kt
- Modify: feature/bookmarks/src/main/kotlin/lava/forum/bookmarks/BookmarksViewModel.kt
- Modify: feature/bookmarks/src/main/kotlin/lava/forum/bookmarks/BookmarksAction.kt

### Steps:

1. Add `SyncNowClick` action to both FavoritesAction and BookmarksAction
2. Add `isSyncing: Boolean` to both states
3. In ViewModels, `SyncNowClick` triggers refresh + sync use cases via `viewModelScope`, toggles `isSyncing`

4. In Screens, add a small sync icon button in the top bar (next to back button or as a trailing icon). Shows spinner when syncing
  5. Compile, spotless, commit
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## **Task 4: Create SyncHistoryWorker + SetHistorySyncPeriodUseCase**

### **Files:**

- Create: core/work/impl/.../workers/SyncHistoryWorker.kt
- Create: core/domain/.../SetHistorySyncPeriodUseCase.kt
- Modify: core/work/api/.../BackgroundService.kt
- Modify: core/work/impl/.../WorkBackgroundService.kt

### **Steps:**

1. Add syncHistory(period) and syncCredentials(period) to BackgroundService interface
  2. In WorkBackgroundService, implement both methods (schedule/cancel PeriodicWorkRequest)
  3. Create SyncHistoryWorker as @HiltWorker — currently a no-op (history is local-only, worker placeholder for future backend)
  4. Create SetHistorySyncPeriodUseCase — persists setting + schedules worker
  5. Compile, spotless, commit
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## **Task 5: Create SyncCredentialsWorker + SetCredentialsSyncPeriodUseCase**

### **Files:**

- Create: core/work/impl/.../workers/SyncCredentialsWorker.kt
- Create: core/domain/.../SetCredentialsSyncPeriodUseCase.kt

### **Steps:**

1. Create SyncCredentialsWorker as @HiltWorker — iterates providers via LavaTrackerSdk, calls checkAuth() for each, reports failures via NotificationService
  2. Create SetCredentialsSyncPeriodUseCase — persists setting + schedules worker
  3. Compile, spotless, commit
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## **Task 6: Expand Menu sync settings UI**

### **Files:**

- Modify: feature/menu/src/main/kotlin/lava/menu/MenuScreen.kt

- Modify: `feature/menu/src/main/kotlin/lava/menu/MenuViewModel.kt`
- Modify: `feature/menu/src/main/kotlin/lava/menu/MenuAction.kt`

### **Steps:**

1. Add `SetHistorySyncPeriod` and `SetCredentialsSyncPeriod` actions
  2. Add `historySyncPeriod` and `credentialsSyncPeriod` to `MenuState`
  3. Wire `SetHistorySyncPeriodUseCase` and `SetCredentialsSyncPeriodUseCase` into `ViewModel`
  4. In `MenuScreen`, expand the sync section from 2 to 4 rows: Favorites, Bookmarks, History, Credentials — each with sync period dropdown
  5. Compile, spotless, commit
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### **Task 7: CI gate + CONTINUATION.md**

Run spotless, constitution check, Go tests, update docs, push to mirrors.